Proposal (Track C, implementation-based): Does Robust Training Protect Trees from Poisoning?

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Abstract

Chen et al. train decision trees to resist small test-time changes to their inputs. Shafahi et al. show that a few correctly labeled poisoned training images can make a model misclassify one chosen test image. We ask: is a tree that resists the first kind of attack also harder to poison? We train normal and robust XGBoost trees on MNIST 2 vs. 6, poison them with 1 to 50 images per target, and compare how often the attack works. We expect robust training to help against test-time attacks but not against poisoning.

Keywords: Implementation-based project (two main papers), Track C (Attacker versus defender), robust decision trees, clean-label poisoning

Team and track. Avinash and Thomas; Track C; **[Implementation]:** [2] main papers (Avinash reads [Robust Decision Trees Against Adversarial Examples], Thomas [Poison Frogs! Targeted Clean-Label Poisoning Attacks on Neural Networks]).

Papers and why related. [Robust Decision Trees Against Adversarial Examples]shows that trees are easily fooled by tiny input perturbations, and that choosing splits by worst-case score hardens them. [Poison Frogs! Targeted Clean-Label Poisoning Attacks on Neural Networks] shows that a few correctly labeled poisons can flip a chosen test prediction without hurting overall accuracy. Since the first is a test-time defense and the second a training-time attack, this project tests whether the defense covers the attack. Trees have no learned feature space, so we adapt the attack: each poison is a base-class image blended with the target at opacity $\gamma = 0.2$ and labeled as the base class.

Question. For robust-training strength $\varepsilon \in \{0, 0.1, 0.2, 0.3\}$ ($\varepsilon = 0$ is the normal tree) and $k \in \{1, 5, 10, 25, 50\}$ poisons per target, what fraction of 50 random targets get flipped to the wrong class? Is it lower, the same, or higher for robust trees? We also report test accuracy and how much input change is needed to fool each model.

Plan.

Now–Oct 22: Read both papers. Set up Python and XGBoost, starting from the authors' RobustTrees code. Train normal and robust trees on MNIST 2 vs. 6 and confirm robust trees are harder to fool.

To Nov 5 (midterm report, 2–3 pages): Experiment set 1: compare normal and robust trees across ε with several random seeds, measuring accuracy and how much input change it takes to fool them. Run one poisoning sanity check. Midterm deliverable: a figure and a table with results for the robust-trees paper, plus difficulties and the plan for the rest.

Nov 6–Nov 25: Experiment set 2: the poisoning study across ε and $k \in \{1, 5, 10, 25, 50\}$ poisons per target, over 50 random targets and several seeds. Also compare poisons built from one base image versus many different ones.

Nov 26–Dec 14: Analysis and write-up; talks on Dec 8 and Dec 10; final report (5–6 pages, excluding references) and code due Dec 14.

Course connections. Lectures [5-12] (Decison Trees, and Classification) ,[22] (Poisoning Attacks).

Why this works. MNIST 2 vs. 6 is small, XGBoost trains quickly, and the robust-tree code is public. Either result is interesting as a defense that doesn't help, or even hurts, against a different attack is worth knowing about.

References. [174] H. Chen, H. Zhang, D. Boning, C.-J. Hsieh. Robust decision trees against adversarial examples. ICML 2019. [171] A. Shafahi, W. R. Huang, M. Najibi, O. Suciu, C. Studer, T. Dumitras, T. Goldstein. Poison frogs! Targeted clean-label poisoning attacks on neural networks. NeurIPS 2018.

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